

Started on	Thursday, 14 May 2026, 1:10 PM
State	Finished
Completed on	Thursday, 14 May 2026, 1:16 PM
Time taken	5 mins 36 secs
Grade	15.00 out of 15.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

_____ sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

- ☐ a. Insertion
- ☐ b. Complexity
- ☒ c. Bubble ✓
- ☐ d. Selection

Question 2 | Correct Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- ☒ a. Linear search ✓
- ☐ b. Merge search
- ☐ c. Bubble search
- ☐ d. Quick search

Question 3 | Correct Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- ☐ a. Binary search
- ☒ b. Linear search ✓
- ☐ c. Hashing
- ☐ d. Jump search

Question 4 | Correct Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- ☐ a. It always starts from the beginning of the list
- ☐ b. It compares elements sequentially
- ☒ c. It can be applied only if the list is sorted ✓
- ☐ d. It works on unsorted lists

Question 5 | Correct Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- ☐ a. Hash search
- ☐ b. Binary search
- ☒ c. Linear search ✓
- ☐ d. Interpolation search

Question 6 | Correct Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

- ☐ a. The first element is returned
- ☐ b. The last element is returned
- ☒ c. The search is considered unsuccessful ✓
- ☐ d. An error is raised

Question 7 | Correct Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- ☐ a. Linear search is more efficient than binary search on large lists.
- ☐ b. Linear search requires the list to be sorted.
- ☐ c. Linear search divides the list into halves.
- ☒ d. Linear search can be applied to both sorted and unsorted lists. ✓

Question 8 | Correct Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- ☐ a. When the list is unsorted
- ☐ b. When the list is very small
- ☐ c. When the list contains duplicate elements
- ☒ d. When the list is sorted ✓

Question 9 | Correct Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- ☐ a. $O(1)$
- ☒ b. $O(\log n)$ ✓
- ☐ c. $O(n)$
- ☐ d. $O(n \log n)$

Question 10 | Correct Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- ☐ a. When the list is sorted
- ☐ b. When the search algorithm finishes
- ☐ c. When the list contains duplicate elements
- ☒ d. When the element is found in the list ✓

Question 11 | Correct Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- ☒ a. The search ends successfully ✓
- ☐ b. The list is sorted
- ☐ c. The search continues in the left sublist
- ☐ d. The search continues in the right sublist

Question 12 | Correct Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n ?

- ☐ a. $\log n$
- ☐ b. $n/2$
- ☒ c. n ✓
- ☐ d. 1

Question 13 | Correct Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- ☐ a. It can only be applied to large lists
- ☐ b. It is slower than linear search for small lists
- ☒ c. It requires the list to be sorted ✓
- ☐ d. It does not work with negative numbers

Question 14 | Correct Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

- ☒ a. Searching ✓
- ☐ b. Finding
- ☐ c. Mining
- ☐ d. Discovering

Question 15 | Correct Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- ☐ a. Binary search works on unsorted lists
- ☒ b. Binary search has a lower time complexity on large, sorted lists ✓
- ☐ c. Binary search can find multiple instances of the target element
- ☐ d. Binary search does not require dividing the list